Week 3: Probability for Pattern Recognition

Week of 22 October 2026 · Videos ≈ 90 min · Lab video ≈ 45 min

This week in one sentence. Describe uncertainty with numbers: probabilities from counts, Bayes' rule, and densities for continuous features, ready for the Bayes decision of Week 4.

Learning outcomes: T3 · L1 (see the syllabus).

What to do, in this order

	Step	Notes	Time
☐	`W03_Math_Toolkit.pdf`, then the **Week 3 Math check** on Moodle (no grade)	Toolkit	20 min
☐	**Video 3.1** Probability as a long-run fraction	§3.1–3.2	17 min
☐	**Video 3.2** Bayes' rule: turning a conditional around	§3.3	15 min
☐	**Video 3.3** Discrete random variables and the PMF	§3.4	11 min
☐	**Video 3.4** Continuous features: from histogram to density	§3.5	24 min
☐	**Video 3.5** The normal density	§3.6–3.8	22 min
☐	Read the Notes again, and do *Check yourself* at the end	all	40 min
☐	**Lab video** and `Lab_W03.ipynb` (no grade)	—	≈ 2 h
☐	`Assignment_W03.ipynb` (**graded, about 1%**) **submit by Wed 28 Oct, 23:59**	—	≈ 1.5 h

How to submit the assignment

1. In Colab: *Runtime* → *Restart and run all*, so every cell shows its output.
2. *File* → *Download* → *Download .ipynb*.
3. Upload the file to *Assignment 3* on Moodle and press *Submit*.

Marking: 1 = runs, every task attempted, every *explain in your own words* answered · 0.5 = more than half attempted · 0 = not submitted, empty or copied. Late work: up to 48 hours without penalty. In the in-person sessions you may be asked to explain your work.

Files and help

Files:
`W03_Notes.pdf` (**the study material; the exam covers what is in the Notes**) · `W03_Math_Toolkit.pdf` · the folder `W03_Slides_web`: slides and live pages of the five videos (open `index.html`) · `Lab_W03.ipynb` · `Assignment_W03.ipynb` · `penguins.csv` (data for both notebooks; in Colab they download it themselves) · interactive page *Density Explorer* (download and open in your browser; works offline).

Help: ask on the course Discord server or by e-mail. **In-person session this week: female section, Thursday 22 October, 12:00–14:00.** Bring your Assignment 2: some of you will be asked to explain it for two minutes. The male section meets next on Thursday 29 October.